

Interactive Visualization of Donald trump vs Hillary Clinton tweet-based sentiment analysis

Sandeep kumar Ramamoorthy

2101119

A thesis submitted for the degree of Master of Science in Artificial Intelligence

Supervisor: Dr. Jon chamberlain
School of Computer Science and Electronic Engineering
University of Essex

August 2022

Table of contents

1. Introduction.....	4
2. Sentiment Analysis.....	5
2.1VADER regressor.....	6
2.2 TF-IDF.....	7
2.3 Word to vector.....	7
2.4 N-Gram.....	8
3. Background.....	8
4. Related work.....	8
5. Data collection.....	9
6. Literature review.....	10
7. Methodology.....	12
7.0VADER sentiment analysis.....	12
7.1NLTK.....	13
7.2Tokenization.....	13
7.3Lemmatization.....	14
7.4TF-IDF vectorizer.....	15
7.5SVR.....	15
7.6Linear Regression.....	16
7.7KNeighbors regressor.....	17
7.8LASSO CV.....	18
8. Tool used.....	18
9. Natural Language toolkit.....	19
10. proposed method.....	19
10.1pre-processing data.....	19
10.2Analysing data.....	19
11 Donald trump sentiment Analysis.....	20

12. Hillary Clinton sentiment Analysis.....	22
13. Experimental Result.....	24
14. Conclusion and Future enhancement.....	28

Abstract

Twitter is a popular social media platform. During the US presidential election, many people express their positive, negative, and neutral opinions about a presidential candidate. The focus of this project is to calculate the sentiment expressed by these tweets. We used a machine learning algorithm to calculate the sentiment of presidential candidate tweets. VADER sentiment analysis and machine learning algorithms like SVR (support vector regression), KNeighbors regressor, Linear regression, and LASSO CV. LASSO CV gave the best accuracy. These are the machine learning techniques used to predict sentiment analysis. We have given a future enhancement of the sentimental analysis.

1. Introduction

Social media platform like Twitter is the most popular and strongest social media in the world. People share news and comments about a leader and a person at world events. Twitter had more than 350 million active users and people used to create around 500 million tweets a day. The researcher's interest has grown due to the rise in Twitter use for political campaigning during elections. Predicting the presidential candidate using Twitter tweets is becoming a popular area. People on Twitter express their positive, negative, and neutral opinion about a presidential candidate during the election campaign. On Twitter, people use a hashtag to indicate the topic of the tweet. This hashtag shows the main content of the topic on Twitter and makes it easy to search for tweets dealing with a specific subject. [1]

The US election was an intense competitive election in the world. Two political candidates like Donald Trump and Hillary Clinton were competing. Donald Trump represents the republic party and Hillary Clinton represents the democratic party. The user conversations on online social networks like Twitter reflect the political tension between these two contenders. The debates among users on online social networks like Twitter reflected the political conflict between these two candidates. Some political-related studies have utilized Twitter data to forecast the political candidate's popularity and support. We can get more exact public opinion about political candidates on Twitter. We have used sentiment analysis on Twitter to observe the attitude, feelings, emotions, opinions, and views of the text using natural language processing (NLP). [2]

The sentiment analysis is the evaluation of the opinion of the people on Twitter. Sentiment analysis is used to classify the text into types of categories such as "positive", "negative", and "neutral". Evaluating an election result using sentiment analysis is extremely difficult. Vader sentiment has been used in this project for the purpose of prediction in people's opinion through positive and negative tweets in twitter. And also, machine learning algorithms like support vector regression, LASSO CV, and Linear regression, K Neighbors regressor has been used in this project. [1]

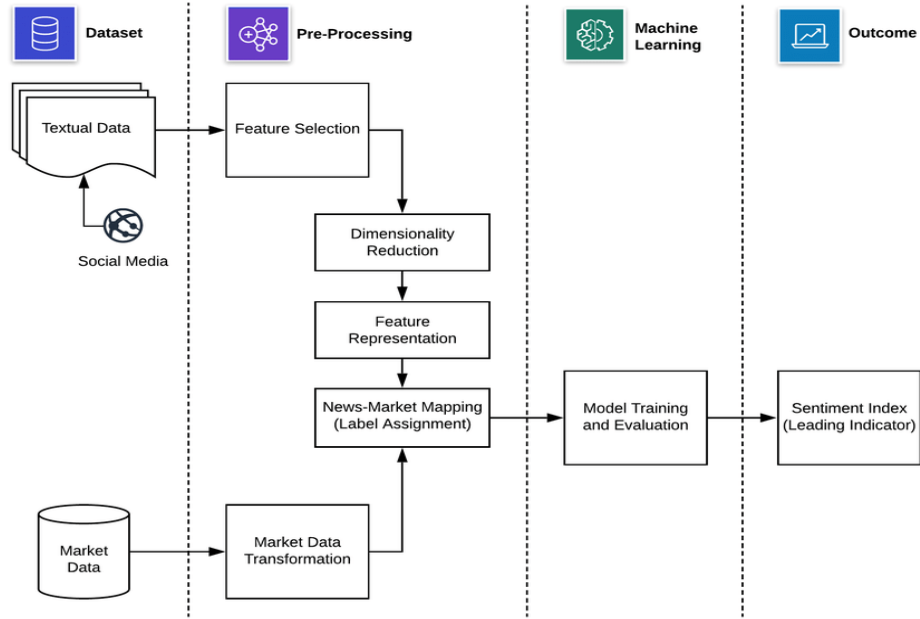
The various methods of sentiment analysis have been used in this project: A machine learning-based method and a VADER-sentimental analysis, and a text vectorization method. As part of the sentiment analysis, tweet text is transformed into a sentiment score that reflects the emotion of the tweet. Such a task is appropriate for VADER sentiment analysis.

Natural language processing. techniques like tokenization, vectorization, and lemmatization. Vectorization is a good approach to converting the input data from the text into vectors of real numbers of machine learning model support. Vectorization is the feature extraction method, by translating text to numerical vectors. The process of converting a word into a number is called vectorization. And the lemmatization is grouping different forms of the same word. It is an important method of natural language processing. Lemmatization is used to group the same word together to form a single word by the word Lemma or dictionary form. Lemmatization is more complex than stemming but it gives meaningful word. Tokenization is the process of splitting a text, or string into a list of tokens. One can think of a token as a word is a token in a sentence, and a sentence is a token in a paragraph.

2. Sentiment Analysis

The sentiment analysis is the process of determining whether the piece word is positive, negative, or neutral on Twitter. It is also known as opinion mining, which derives the opinion of the people on Twitter. The sentiment analysis can be used in the marketing field, business, and political fields. In the business field sentiment analysis is used to develop strategies, to understand the customer feeling toward the brand in the shop. In politics, sentiment analysis is used to keep track of political views like detecting people's views on political candidates using Twitter tweets. The sentiment analysis performs actions in various fields like business, shops, and political election campaigns. The sentiment analysis can be more helpful in the following ways.

- the main process of getting more understanding of customer opinion towards the brand in shop.
- Obtain knowledge that will enable you to make your products and services better.
- It monitor's the brand reputation in real-time.
- It will make the business more responsible and profitable to customer feedback.
- It keeps customers happy and gives profit by detecting people's opinions.
- Sentiment analysis gives a good understanding of people's opinions about political candidates.
- Sentiment analysis reacts quickly to negative sentiment and turns It around.



Process of Sentiment Analysis FIG 1 [24]

On Twitter, by detecting people's opinions, feelings assigns sentiment score. For instance, negative to positive sentiment gives a score from 0 to 10. Depending on the score we can understand people's feelings. The quantity, kind, and intensity of the emotions expressed, as well as the context in which they were expressed, are all criteria that go into calculating a sentiment score. Sentiment ratings can be beneficial for several things, including figuring out customer satisfaction or figuring out whether a text is positive or negative. The sentiment analysis methods are discussed below.

2.1 VADER sentiment

The Vader represents the valence award dictionary and sentiment reasoner. Vader sentiment is used for text sentimental analysis. Which can handle people's opinions, feelings, words, emojis, and slang found on social media. Vader sentiment is faster than the machine learning algorithm [8]. The Vader sentiment is used to give positive and negative score to the people's opinion. The Vader sentiment gives positive, negative, and neutral score to the people's opinion in the twitter. The sentiment values for the negative, neutral, and positive score ranges from 0 to 1. And it evaluates the people negative emotions from -1 to positive emotions to 1. So, the Vader sentiment analysis is used to evaluates the people's opinion through the people's positive and negative text.

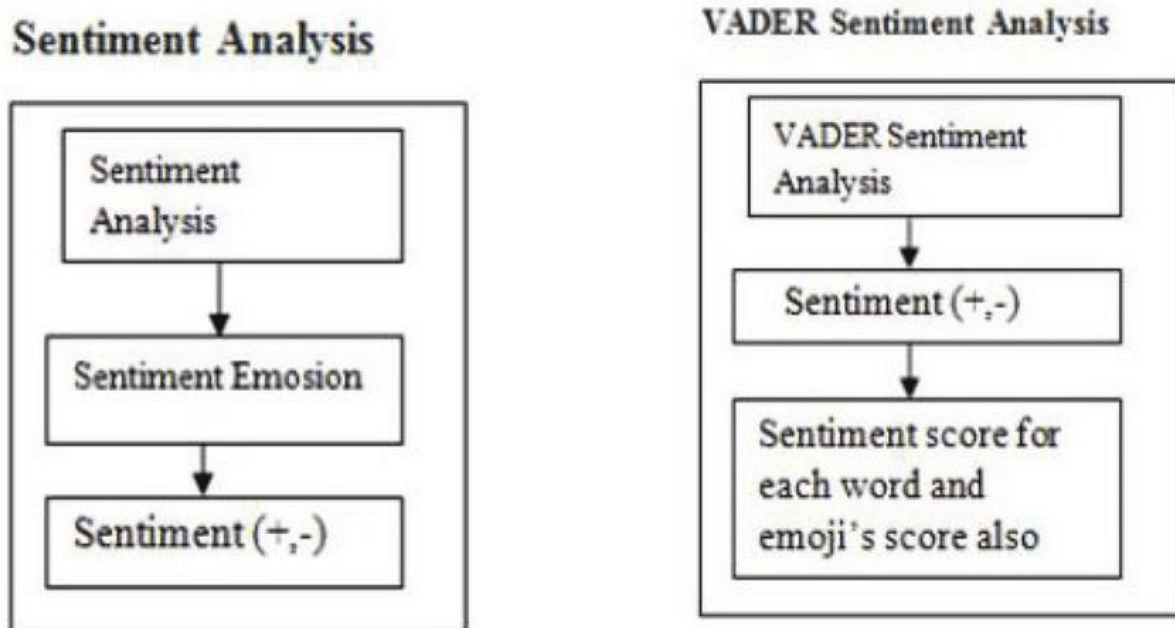


FIG 2 [9]

2.2 TF-IDF

The TF-IDF is term frequency and inverse document frequency. It is one of the simple methods to understand the context of the text. It is used to find the related content and important text in the larger text. The term frequency gives us the number of words that appears in each document in the corpus. It measures how frequently a word appears in a document relative to all the other words in that document. Its increases the number of occurrences of the word with the documented increase. TF is unique to each document.

2.3 Word to vector (word2vec)

Word to vector is a group of models which produce word embeddings. It is determining a word and its surrounding words. The word-to vector uses a machine learning algorithm to learn word associates from a large corpus of text. The design of embedding words is a vector of real-valued values. These vectors maintain both the semantic regularity between word pairs and the syntactic regularity by making related words have similar vectors. Word2vec in the beginning creates high-quality word vectors more quickly than with other methods. [8]

2.4 N-Gram

N-Gram is the language modelling. It is used to determine the probability of given n-grams within sequence of words. When addressing negations, sentiment analysis may be more accurate if a sentiment dictionary uses larger n-grams. The best N-Gram models predict the next word in the sentence. And the grouping of the words is called trigrams in N-Gram.

3. Background

Social media is the strongest platform where People can express their opinions, feelings, and emotions about political candidates on Twitter. Twitter currently has more than 500 million active users and receives more than 600 million tweets per day. Sentiment analysis is the part used to determine the people's opinion about a political candidate using tweets. Performing sentiment analysis on the tweets is one technique to determine whether a contender has a chance of winning the election.

Twitter was a tool that Barack Obama utilized to advertise his presidential run in 2008 [4]. The administration of Obama's online presence was handled by 100 employees of his campaign. The election outcomes were greatly impacted by this. Since then, parties hire research associates to undertake electoral manifesto and public opinion analyses before every election. They can learn what the preferences of their constituents for the next president are in this way. People show their opinions and feeling about the election candidate. To express their viewpoints, people from all around the world tweet about the US presidential candidates and elections. From Twitter, using sentimental analysis determine the people's view of a political candidate. The top ten political trending topics on Twitter on August 12, 2020, included both Republican nominee Donald Trump and Democratic nominee Joe Biden. [5]

VADER sentimental analysis and machine learning algorithms like SVR (support vector regressor), KNeighbor regressor, Linear regression, and LASSO CV. These techniques have been used to determine the sentiment analysis in Donald Trump and Hillary Clinton's political election.

4. Related work

The most widely used language on earth is English. The majority of sentiment analysis research is done in English. A. Tumasjan, T. O. Sprenger, P. G. Sandner, and I. M. Welpé conducted a prediction study using Twitter in 2009 and discovered that 50% of users only tweeted once in their corpus, while the remaining 50% produced 90% of the tweets. They used a data set of approximately 100K tweets, which is too unclear to predict for a country with a huge population. [6]

In the paper “A novel classification approach based on Naïve Bayes for Twitter sentiment analysis”, To predict the emotion of a text using Naive Bayes classification, they employed a positive and negative terms file. However, there is a potential of producing a biased result because the ratio of positive to negative terms is not always equal. They suggested two approaches to resolve this problem. The first method involves counting how many positive and negative words there are when

calculating weights, while the second method involves finding significant terms that can be used to determine the type of test text. They attained 85.33% accuracy in this manner. [6]

Although it's nearly difficult to have every positive phrase in the positive word file and every negative word in the negative word file, using this model to analyze our tweets dataset may result in lower accuracy and the possibility of many tweets having the incorrect sentiment. [5]

Based on keywords like Donald Trump and Hillary Clinton, B. Joyce and J. Deng collected tweets from the 2016 presidential election. For tweet gathering, they made advantage of Twitter's streaming API. Democratic and Republican were among the terms they used, as well as the entire names of the leading candidates for office, including Donald Trump. They collected almost 79 million tweets over two months from 10,000 different users because they used Twitter's official tweet streaming API. Then, 1.9 million tweets with emojis were removed. The data set decreased further to 783K tweets after eliminating retweets. They identify each tweet's emotional orientation using the Emojis Sentiment Ranking. They did that by removing the emoji from the statement and then transcribing it into Unicode. Then using regex on Unicode, a score was found from a customized list of 522 original emoji characters. This rating will indicate the sentence's emotional content or feeling. They employed a multinomial Naive Bayes classifier to build a sentiment classifier. [5]

Predicting the results based on emojis will not be accurate. Users may be added the emoji as a sarcastic on Twitter. By doing sentiment analysis only word orders in Twitter. To fix this problem we have used Vader sentiment and developed a model to determine the sentiment on word numbers. Machine learning algorithms like support vector regression, linear regression, and KNeighbor regressor, LASSOCV have been used to train a model with word order in the existing work. And these machine learning algorithms seem to be the suitable algorithm for doing sentiment analysis.

5. Data collection

We collected the Twitter tweet dataset from Kaggle on the internet. Twitter is a social network that allows the publication of tweets like short messages. Twitter has more than 350 million active users. The researcher's interest has grown due to the rise in Twitter use for political campaigning during elections. The collection of data is the important stage for sentimental analysis. Most of the political related dataset we can get from twitter. We got Information through social media networks is said to be as social dataset. Social media is the best data sources for the researchers. Through social media we can get many Information. Twitter is the best mobile-based applications that let users to access and to show their feelings through comment. [7]

A brief weblog is made up of paragraphs that express an opinion, information, a personal entry from a journal, or links. Together referred to as a post, which is organized chronologically with the most recent item showing up first, in the manner of a research paper. Blogs are a useful tool for conducting sentiment analysis on many entities.

6. Literature Review

This paper [10] was written by Saad and Yang in 2019. They used machine learning algorithm and ordinal regression model for determining the tweets using sentimental analysis. They collected data and then pre-processed the tweets. The machine learning methods such as SVR (support vector regression), Random Forest, and logistic regression and decision tree was used for the sentimental analysis. Using twitter dataset, the model was tested. The tested model gave good accuracy. Decision tree was the best model which gave good accuracy when compared to other models.

In paper [11] Feizollah and others 2019, have performed a sentiment analysis on Twitter tweets. In this paper, they are going to determine the halal products such as halal tourism and halal cosmetics. They got the dataset through the twitter search function. They used deep learning algorithms like CNN, LSTM, and RNN. In data pre-processing, they converted sentence into words and assigned sentiment values like positive and negative based on text which gives 0 to 1 to the word. Then evaluating the overall sentiment score of tweets. In these prediction models, LSTM and CNN gave the best result.

In this paper [21] Afzaal and others suggested an innovative method of aspect-based sentiment classification in 2019, to help tourist people to find the good hotel and better environment. Classification models and NLP techniques has been used. Experiments has been implemented on the datasets taken from social reviews websites. The outcomes demonstrated that the proposed model was successful in both classification and recognition.

In the paper [20] Mukthar and others in 2018 have proposed a two approach for the sentiment analysis of urdu blogs. Lexicon-based and supervised machine learning approach. The models like SVM, DT, and KNN were used in this experiment. Urdu sentiment Analyzer has been created to perform sentiment analysis. The results of the experiments proved that the Lexicon-based model gave best accuracy.

In a paper [12] written by Mittal S.A 2016. Describes how sentiment analysis is used to manipulate product ratings. The producer can get the proper feedback from a client by promoting the products. Based on the feedback, they change the products some users fail to give their feedback. The purpose of this research is to develop a strategy that may automatically provide feedback based on data gathered from Twitter to eliminate the issue of providing input. They applied the SVM approach and obtained an accuracy score of 80%. This method provides timely and helpful feedback.

The paper [13] written by Shahare F.F in 2017, this paper describes the sentimental analysis of people's opinions related to the high faculty of India. They must provide comparisons between the outcomes of the following machine learning algorithms. The models like SVM, Artificial neural network, and Naive Bayes model. Interestingly since the Naive Bayes model is simple and it is best in finding the small emotional text in larger dataset. as a result, Naive Bayes beats SVM to classify matter polarity. It shows a decent accuracy.

In this paper [19] Ray and Chakrabarti 2019, have developed a deep learning system to extract text features and analyze users' sentiments. A seven-layer deep learning CNN was used for the aspect extraction from text in the opinionated sentence. The combination of deep learning and rule-based techniques is to improve the performance of sentiment score. In the end, the suggested method achieved the best accuracy.

In this paper [14] Zhao and others have proposed a new image-text consistency-driven multi-modal sentiment analysis method, it will be useful for extract information from both text and image. The classic SentiBank model was used to give accurate sentiment analysis. And it is used to develop machine learning method. this model has achieved the best results when compared to other method.

In this paper [15] (Kaur, 2015). Describe the geographic area flood data set that was obtained via Twitter and consider people's perspectives. They classified the information using the Naive Bayes formula, and the results showed a 67% accuracy rate. To address this situation in an incredibly better way, they need to gather various resolutions from the affected people that are helpful for both government and private organizations. These methods are less complex than lexicon method.

In the paper [22] park and others 2019, have created a sentiment-discriminative semi-supervised aim to address the problem by documenting sentiment data. The recommended model secured the local structures in addition to reflecting the incomplete data. Real-time datasets were used to assess the proposed model. The result demonstrated that the proposed paradigm was operating successfully.

In one of the papers [17] Smadi and others 2018, have provided current method to solve the challenges of arabic hotel review. RNN and SVM were implemented. And trained with lexical, word and semantic features. To evaluate the proposed model, the datasets were used. SVM was giving the best accuracy compared to RNN. [17]

In this paper [18] Abdi and others have proposed a method for classifying the user's viewpoint indicated in reviews using deep learning. This model was used to indicate the word embeddings. RNN and LSTM has been used to overcome the challenges in the problem. In one of the papers, park, and others have developed a deep learning model to enhance performance. To improve the performance, two questions have emerged for good performance. The content attention needed to be sophisticated to non-linearly combine several attention outcomes and when referencing complicated phrases, it assumes the entire context. This model delivered the highest level of performance.

In this paper [13] Shahare 2017, relates that describing the sentimental analysis of report information on social media. Naïve bayed and Levenshtein formula has been used to categorize the sensation into whole separate groupings from news data from social media. This technique provides higher performance. And it provides better performance in terms of accuracy. As result, the Levenshtein formula performs the best text process on the information. It operates quickly and offers the highest degree of precision in the process of a lot of information.

7. Methodology

In methodology, we have collected data from Twitter. For training, we used 80 percent of the data, and the rest of the data are used for testing purposes. We have used techniques for sentiment analysis are machine learning algorithms like SVR (support vector regression), linear regression, KNeighbors

regressor, LASSOCV, And VADER sentiment, and NLTK like lemmatization, tokenization, and TF-IDF vectorizer.

7.0 VADER Sentiment Analyser

The VADER is the valence-aware dictionary and sentiment reasoner. VADER is the rule-based sentiment analysis. Which is used to evaluate the people's opinion on social media. It contains the list of lexical features. Which are generally labelled as positive and negative semantic orientation. In social media, people can share and have a conversation with different people and to post messages in text form. The vader is used to classify the people opinion into positive and negative. The vader will assign score to the text based on the people feelings and emotions. It will give the good polarity of the tweets.

The work logs are first segmented into sentences, cleaned by eliminating stop words and salutations, and then steamed. In any language, in repeated use of words, stop words has been used. Stop words are mainly concentrated on the keywords. Eliminating unnecessary words in the process. The dimensionality of term space is decreased by eliminating stop words. The act of reducing a word into single word to its stem or root form is known as stemming. Words like connect, connected, connecting, and connections, for instance, can all be shortened to "connect." This method's goal is to eliminate suffixes from stems so that they precisely match, to save time and memory space. Either language dictionaries or algorithms provide the foundation for stemming. Each phrase is provided to the VADER Sentiment Analyzer function after the first data pre-processing processes have been done to the data to extract the score and polarity of each word in the sentence. The top negative terms in the studied work logs are then found using the outcome. [26]

7.1 NLTK (Natural language toolkit)

The NLTK is the natural language toolkit. It is a leading program to work with human language data. All the NLP techniques had been used in this method. It is the collection of all NLP method. This method is used in many fields like companies to get insight from unstructured data. NLTK is the process of using software or a computer to manipulate or comprehend voice or text. Human interaction, understanding of one another's viewpoints, and providing the proper response are examples of an analogy. In NLP, a machine performs this interaction, comprehension, and response in place of a person.

PROCESS OF NLTK

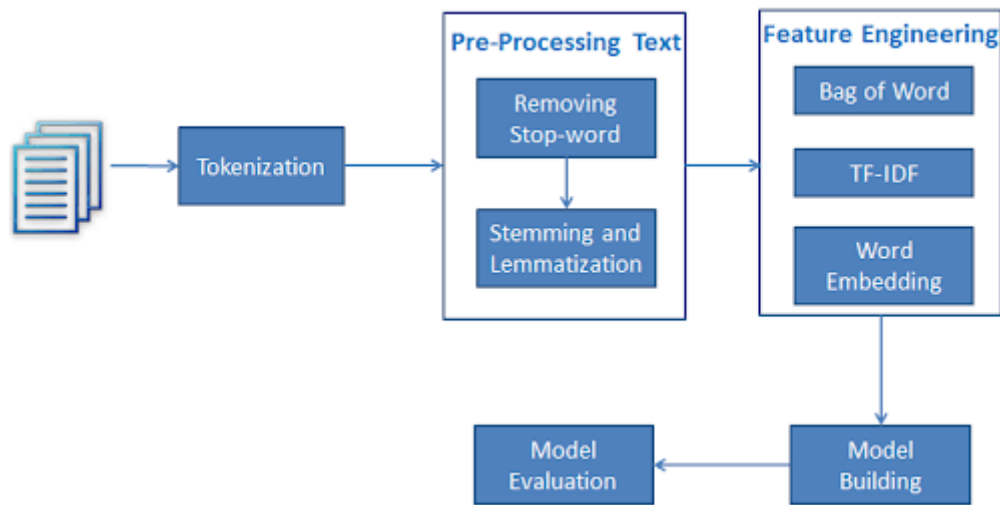


FIG 3

[27]

7.2 Tokenization

The tokenization is the minimal unit that the machine can understand and process. Without tokenization, the text string cannot be processed. And it is the process of splitting the raw string into token vectors. Depending on the requirements of the NLP application and the complexity of the language itself, tokenization complexity varies. The word tokenization is a generic and robust method of tokenization in the text corpus. The word tokenization comes from NLTK. Tokenization is an important method of natural language where the raw text is often processed.

It is referred to as "Word Tokenization" if the text is divided into words, and "Sentence Tokenization" if it is divided into sentences. For word tokenization, the character "space" is often used, while for sentence tokenization, characters like "periods," "exclamation points," and "newlines" are typically employed. Depending on the work at hand, we must select the best methodology. A few characters, such as spaces and punctuation, are disregarded throughout the tokenization process and are not included in the final list of tokens. [28]

PROCESS OF TOKENIZATION

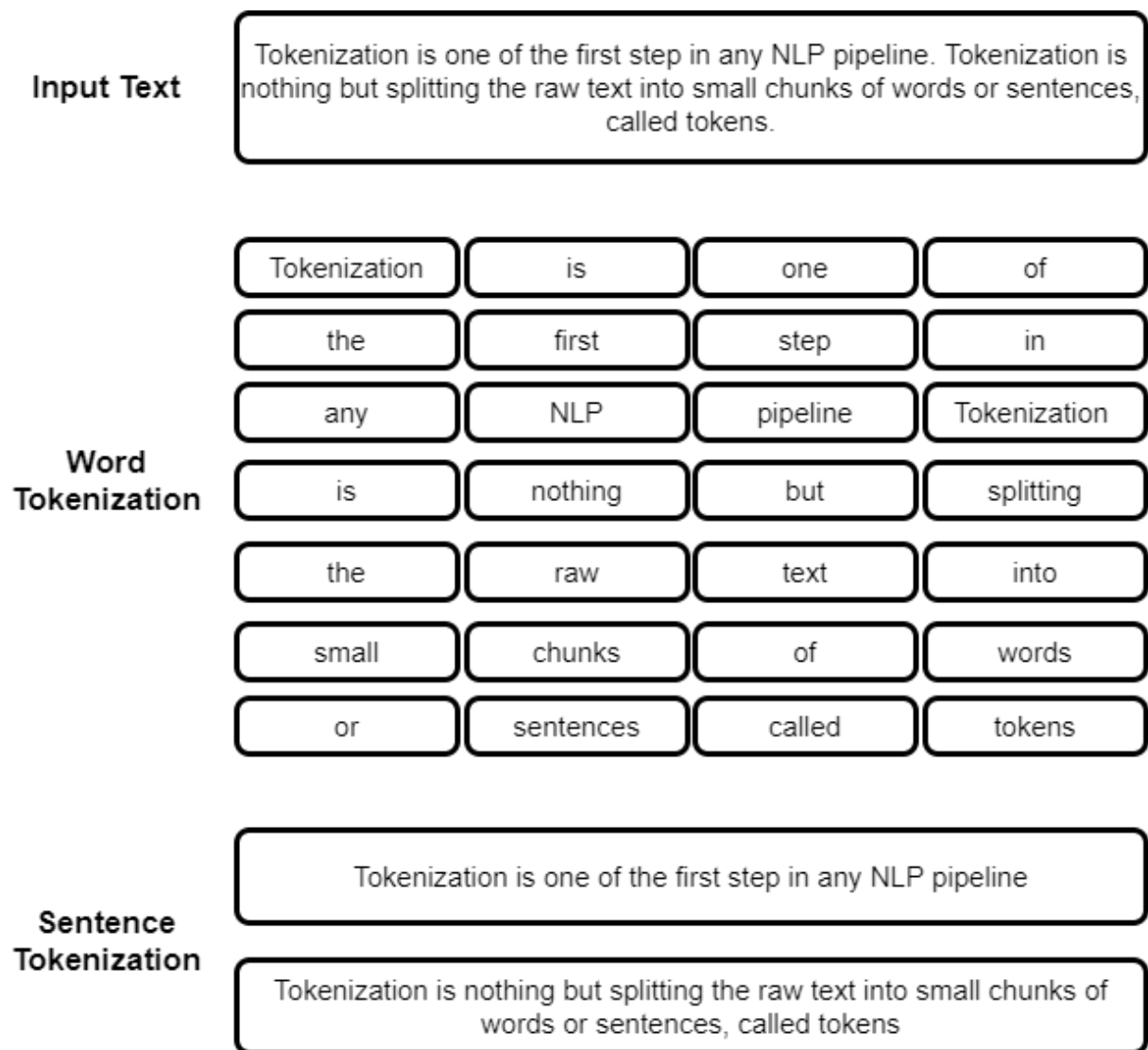


Fig 4

[28]

7.3 Lemmatization

Lemmatization is the process of grouping together different forms of the same word. It is an important method of natural language understanding. Lemmatization allows the user to query a version of the base word to get the relevant result. Due to the usage of lemmatization in search engine algorithms, users are allowed to query any word's inflectional form and receive relevant results. And the lemmatization is more closely related and complex than stemming. It's used in comprehensive retrieval systems like a search engine. And it's used in compact indexing. Linguistics is the key to this

technique. It is required to look at each word's morphological analysis to obtain the correct lemma. To perform that type of analysis, dictionaries for every language are necessary.

7.4 TF-IDF VECTORIZER

The TF-IDF is the term frequency and inverse document frequency. It transforms text to feature vectors. And it is the process of transforming the text into numbers for the purpose of evaluation. Which is used for the prediction using machine learning models. The term frequency gives us the number of words that appears in each document in the corpus. It measures how frequently a word appears in a document relative to all the other words in that document. Its increases the number of occurrences of the word with the documented increase. TF is unique to each document.

7.5 Support vector regression (SVR)

The SVR is the support vector regression in the supervised machine learning algorithm. which is used to predict discrete values. It is quite like the support vector machines. The main principle of SVR is to work with continuous values. This method is good in minimising the error rate. SVR and SVM are bit different. In that SVR is a regressor used to predict continuous ordered variables, whereas SVM is a classifier used to predict discrete categorical labels.

The SVR is known for reducing the error rates and it will work in continuous value. It will follow the principles of SVM. Using SVR, trying to fit the line within threshold. In given data points, SVR locate the curve to match the vector point to curve. The Support vector is best to Finding the closest match between the data points and the function. The required actions to make SVR function properly are gathering the training materials, choosing a kernel, its parameters, and correlation matrix creation. [29]

SUPPORT VECTOR REGRESSION

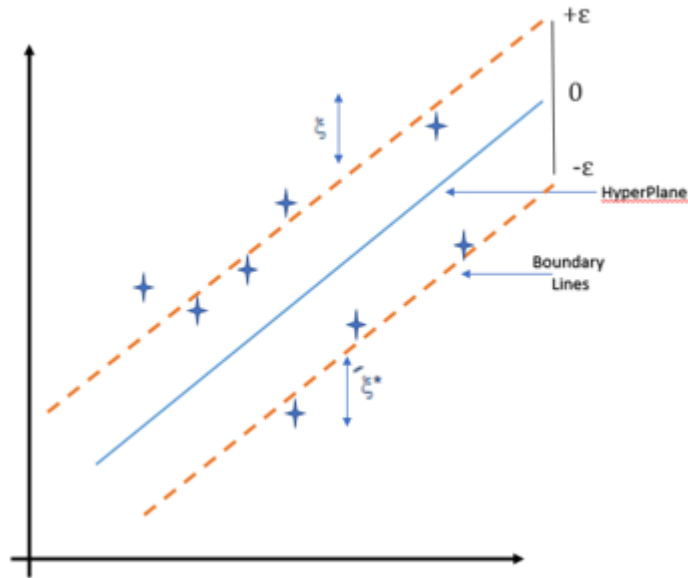


FIG 5

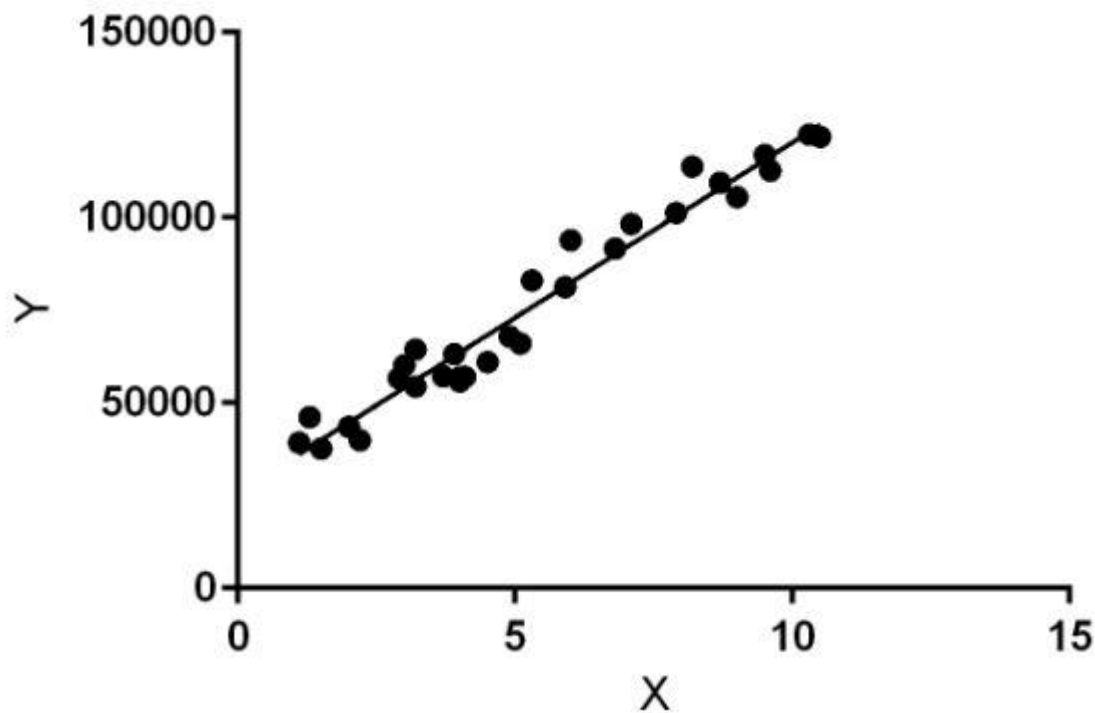
[29]

7.6 Linear regression

linear regression is one of the best methods in machine learning algorithms. It is used for the prediction in variable value on the value of another variable. It helps in predicting the dependent variable value. It fits the relationship between two variables using best fitting straight line called a regression line. X which is called independent variable predicts the value of dependent variable. Which is represented as Y.

The linear regression estimates the fitness between actual and expected output values by fitting straight line. Using straight forward linear regression with least square technique, the best fitness will be found. This method is a simple algorithm and it's easy to interpret mathematical formulas. It can be applied to various fields like business and education. And it is supervised learning, it performs the regression task.

Based on the independent variable, linear regression performs the prediction task on a dependent variable. This technique will forecast the linear relationship between input and output. Linear regression belongs to both statistics and machine learning. The learning algorithm is used to create a linear regression model. Making the most accurate predictions possible or minimizing model error are the main goals of machine learning, the linear regression is best in the field of predictive modelling. [30]



LINEAR REGRESSION graph FIG 6 [30]

7.7 KNeighbors regressor

The KNeighbors regressor is a supervised learning algorithm. it is used for both regression and classification. It is a simple algorithm that stores all cases and predicts the numerical target. KNN is used in statistical and pattern recognition. By calculating the distance between the test data and all of the training points, KNN tries to predict the proper class for the test data. Then choose the K spots that are closest to the test data. KNN regressor is bit similar to the KNN classifier.

Only continuous variables can be used with the three-distance metrics mentioned above. The Hamming distance, which is a measurement of the number of instances in which similar symbols are different in two strings of identical length, must be used in the case of categorical variables. The target value of the closest neighbor is all that is predicted when only one neighbor is used.

The KNN technique may be applied to issues involving classification and regression. To predict the values of any new data points, the KNN algorithm takes advantage of "feature similarity." It is a non-parametric method. It approximates the independent variables and continuous result by averaging the data in the same neighborhood. The analyst must decide on the neighborhood's size, or cross-validation can be used to determine the size that minimizes the mean-squared error. The KNN model is easy to implement and handles non-linearities well. Fitting the model in KNN is also quick. For a larger dataset, KNN takes a long time to fit. [31]

7.8 Lasso CV

The Lasso CV is the least absolute shrinkage and selection, operator. It is a regression analysis method. It is a statistical formula for the regularization of data models and feature selection. Regularization is an important method in LASSO. The regularization is used to avoid the overfitting of the data when the trained and tested data are much varying. The LASSO regularization technique is classified into two types: they are ridge regression and LASSO regulation.

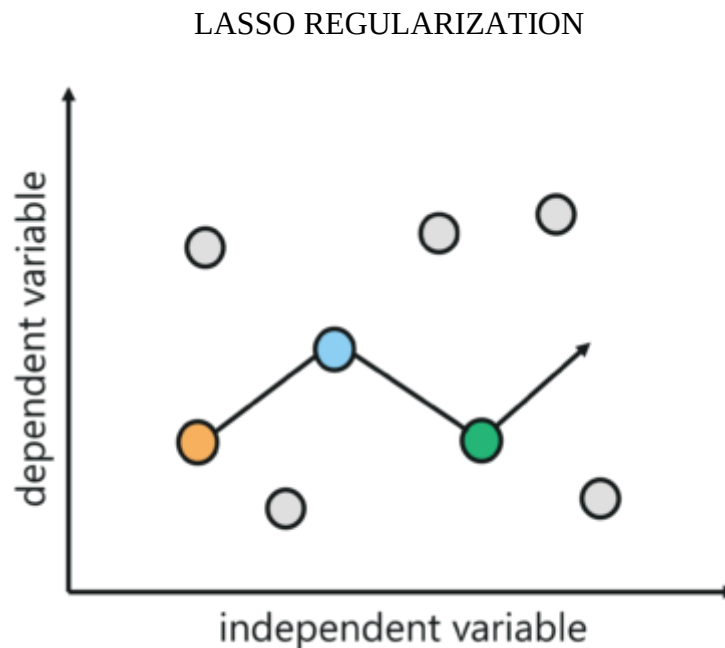


FIG 7

[32]

The LASSO regression is different from ridge regression. It uses coefficient values for normalization. When we have correlated variables, the fundamental issue with lasso regression is that it only keeps one variable and puts the other connected variables to zero. That may cause some information to be lost, which would give lower accuracy in our model. To get accurate prediction, regularization technique has been used. Lasso regression uses shrinkage, which involves drawing data values closer to a central point, such as the mean value. The term "lasso regression" refers to the L1 regularisation method. The term "ridge regression" refers to the L2 regularisation method. [32]

8. Tool used

There are many methods and algorithms used for sentiment analysis. This project was worked in Jupyter notebook. And the report was written in word.

DATASETS: The datasets are collected from Kaggle in the internet.

OPERATING SYSTEM: windows operating system have been used.

MACHINE LEARNING ALGORITHM: SVR, Linear regression, KNN Regression, LASSO CV. These are the machine learning algorithms that have been used in this project.

SENTIMENT ANALYSIS: VADER sentiment analysis have been used to detect sentiment analysis in Twitter tweets.

9. Natural language Toolkit

NLTK is the best platform to work for building python programs with data. Along with a collection of NLP techniques. This method is used in sentiment analysis for the purpose of sentiment analysis by using collection of methods in the NLTK.

10. Proposed method

In this method, this project consists of three steps. The first step is to gather Twitter data. And the second step is to pre-process the data to remove the unimportant messages from tweets. The third step deals with the usage of the NLTK VADER sentimental analyzer and the scoring method applied to VADER to evaluate text on Twitter. The purpose of the data acquisition was to obtain Twitter data. And this method was used to extract twitter data for the analysis. The gathered data will be stored in an excel sheet. The tweets are micro-blog posts made to Twitter. There are only 140 characters allowed on Twitter. Most tweets are text-based with embedded URLs, images, usernames, and emoticons. There are also typos in them.

10.1 Pre-processing data

Therefore, pre-processing procedures were used to get rid of pointless information from the tweets. To get the best accuracy, the cleaned data are most suitable for feature extraction and mining. A final sample of some tweets was obtained after the tweets were pre-processed to remove duplicates and retweets from the dataset. To extract each tweet's primary message, each was processed. We processed each tweet to get the main message. And to pre-process the data we used NLTK (natural language toolkit). To find and remove special characters in tweets such as URLs ("http://url"), retweets (RT), user mentions (@), and extra punctuation, a regular expression (Regex) was run in Python. Hashtags (#) are added to tweets to categorize content and are frequently used to give useful information which relates to the topic of the tweet. [23]

The method of NLTK is the process of converting the tweets that were tokenized into individual words or tokens, converted to lowercase, removed stop words (words that don't convey any sense, such as a the, he, them, etc.), and stemmed using the Porter stemmer utilizing a variety of NLTK methods. The dataset was prepared for sentiment analysis after the pre-processing steps are completed.

10.2 Analysing data

In the third step, the opinions in the tweets were classified. VADER sentiment analyzer was applied to the datasets. VADER is a rule-based sentiment analyzer. To classify our dataset, we first built a sentiment intensity analyzer. After that, the emotion was ascertained using the polarity scores approach. The pre-processed tweets were categorized as positive, negative, neutral. The VADER Sentiment

Analyzer is used to determine the people's opinion, and feelings. A good statistic for assessing the mood of a particular tweet is the compound value. The threshold values in the suggested approach are used to classify tweets as either good, negative, or neutral. We developed a evaluating formula method to determine the overall sentiment polarity in each tweet: positive, negative, and neutral. [23]

11. Donald trump tweet sentiment Analysis

Several data cleaning processes like removing the commonly used stop words and the Punkt function of NLTK library build a model to create a model for abbreviation words, colocation, and words. Then the tokenization takes place in the tweets to process the text into splitting a string, text into a list of tokens. And the lemmatizing of the word like grouping the inflected forms of the word to analyze them as a single word. If the form of the word is not lemmatized, then there will be a deviation in the statistical results. The lemmatization is achieved by using NLTK. At last, stopwords have been used to remove the words that we don't want to use in tweets. The stopwords are used to eliminate the unimportant words in the tweet. In the final, we will get the sentimental score for the Donald trump sentimental analysis.

	handle	text	is_retweet	original_author	time	lang	retweet_count	favorite_count	hour	month	day	month_f	sentiment_score
0	realDonaldTrump	Join me for a 3pm rally - tomorrow at the Mid-...	False	NaN	2016-09-27 22:13:24	English	2181	6172	22	9	27	SEP	0.2960
1	realDonaldTrump	Once again, we will have a government of, by a...	False	NaN	2016-09-27 21:08:22	English	4132	11239	21	9	27	SEP	0.2960
2	realDonaldTrump	Hillary Clinton's Campaign Continues To Make F...	False	NaN	2016-09-27 20:14:33	English	3833	9840	20	9	27	SEP	0.0000
3	realDonaldTrump	'CNBC, Time magazine online polls say Donald T...	False	NaN	2016-09-27 20:06:25	English	4236	10948	20	9	27	SEP	0.0000
4	realDonaldTrump	Great afternoon in Little Havana with Hispanic...	False	NaN	2016-09-27 19:59:04	English	4878	14110	19	9	27	SEP	0.8519

FIG 7

After getting a sentimental score for Donald trump. We are using the VADER sentiment analyzer, the Donald trump tweets have been processed in VADER sentimental analysis. VADER can understand the sentiment of text containing emotions, and it works well on Twitter tweets. In this step, we are implementing the sentiment labelling technique to make the predicted sentiment score in a human-readable format. This will help to map the sentiment score to 3 human understandable classes (positive sentiment, neutral sentiment, and negative sentiment), by classifying the numerical values for a threshold. A good statistic for assessing the mood of a particular tweet is the compound value. The threshold values in the suggested approach are used to classify tweets as either good, negative, or neutral.

```

i = 0
vader_sentiment = [ ]
while(i<len(tweets_trump)):
    if ((tweets_trump.iloc[i]['sentiment_score'] >= 0.05)):
        vader_sentiment.append('positive')
        i = i+1
    elif ((tweets_trump.iloc[i]['sentiment_score'] > -0.05) & (tweets_trump.iloc[i]['sentiment_score'] < 0.05)):
        vader_sentiment.append('neutral')
        i = i+1
    elif ((tweets_trump.iloc[i]['sentiment_score'] <= -0.05)):
        vader_sentiment.append('negative')
        i = i+1

```

FIG 8

In the length of the tweets, if the sentiment score is more than 0.05. then it is positive. Or if the score is equal to 0.05. then it is neutral. If the score is below 0.05. Then it is negative. By using this technique, we will get the VADER sentiment score of Donald trump from the tweets.

```

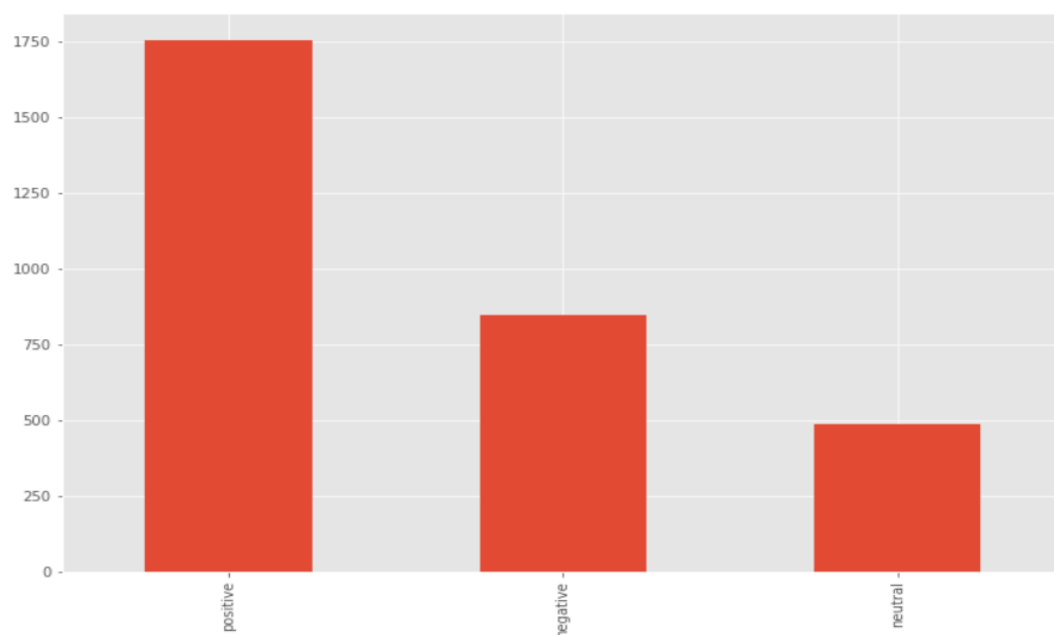
Out[22]: positive    1755
        negative    849
        neutral     489
        Name: vader_sentiment_labels, dtype: int64

```

FIG 9

Then the VADER sentiment visualization of Donald trump's tweet is plotted in the graph chart.

FIG10



12. Hillary Clinton sentiment Analysis

The same process will be done in Hillary Clinton's sentiment analysis. To generate a model for abbreviated words, colocation, and words, many data cleaning techniques were applied, such as deleting frequently used stop words and using the Punkt function of the NLTK library. Then, in the tweets, the tokenization process splits the text into a string and a list of tokens. Additionally, lemmatizing a word involves organizing its inflected forms into a single group for analysis. Then, in the tweets, the tokenization process splits the text into a string and a list of tokens. Additionally, lemmatizing a word involves organizing its inflected forms into a single group for analysis. The statistical findings will deviate if the word form is not lemmatized. Utilizing NLTK, lemmatization is accomplished. The terms that we don't want to use in tweets have finally been removed using stopwords. the stopwords that were utilized to cut out the tweet's unnecessary words. The emotional score for the Hillary Clinton sentimental analysis will be revealed in the end.

Out[26]:

	handle	text	is_retweet	original_author	time	lang	retweet_count	favorite_count	hour	month	day	month_f	sentiment_score
0	HillaryClinton	The question in this election: Who can put the...	False	NaN	2016-09-28 00:22:34	English	218	651	0	9	28	SEP	0.4404
1	HillaryClinton	If we stand together, there's nothing we can't...	False	NaN	2016-09-27 23:08:41	English	916	2542	23	9	27	SEP	-0.4717
2	HillaryClinton	Both candidates were asked about how they'd co...	False	NaN	2016-09-27 22:30:27	English	859	2882	22	9	27	SEP	-0.6597
3	HillaryClinton	This election is too important to sit out. Go ...	False	NaN	2016-09-27 21:35:28	English	1303	2849	21	9	27	SEP	0.4767
4	HillaryClinton	When Donald Trump goes low...register to vote....	False	NaN	2016-09-27 21:25:31	English	1833	4954	21	9	27	SEP	-0.2732

FIG 11

After giving Hillary Clinton an emotional score. The Hillary Clinton tweets have been analyzed using the VADER sentiment analyzer, which we are currently utilizing. When used on tweets from Twitter, VADER can comprehend the sentiment of text that contains emotions. We are using the sentiment labeling approach in this stage to convert the anticipated sentiment score to a format that is understandable by humans. Categorizing the numerical values about a threshold will assist in mapping the sentiment score to three human comprehensible subcategories (positive sentiment, neutral sentiment, and negative sentiment). The compound value is a useful metric for determining how an individual tweet is feeling. The recommended technique uses threshold values to categorize tweets as positive, negative, or neutral.

```
In [27]: i = 0
vader_sentiment = [ ]
while(i<len(tweets_hillary)):
    if ((tweets_hillary.iloc[i]['sentiment_score'] >= 0.05)):
        vader_sentiment.append('positive')
        i = i+1
    elif ((tweets_hillary.iloc[i]['sentiment_score'] > -0.05) & (tweets_hillary.iloc[i]['sentiment_score'] < 0.05)):
        vader_sentiment.append('neutral')
        i = i+1
    elif ((tweets_hillary.iloc[i]['sentiment_score'] <= -0.05)):
        vader_sentiment.append('negative')
        i = i+1
```

FIG 12

If the emotion score for the whole duration of the tweets is more than 0.05, it is considered positive. Alternatively, if the result is 0.05 or less, it is neutral. The result is negative if the score is less than 0.05. Using this method, we can extract Hillary Clinton's VADER emotion score from her tweets.

```
Out[29]: positive    1290  
        negative     736  
        neutral      603  
        Name: vader_sentiment_labels, dtype: int64
```

FIG 13

Then the VADER sentiment analysis visualization of Hillary Clinton's tweet is shown in the graph chart.

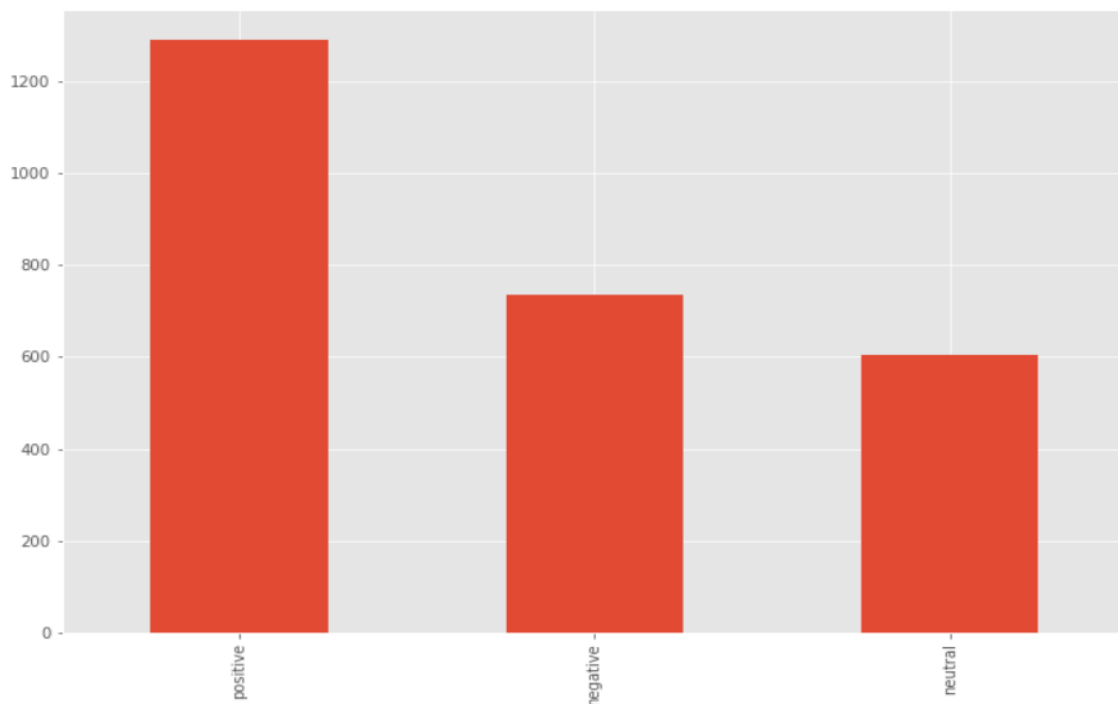


FIG 14

13. Experimental Result

The experimental analysis of Interactive visualization of Donald Trump and Hillary Clinton tweet-based sentiment analysis was shown here. VADER sentiment analysis and machine learning algorithms like SVR, KNneighbors regression, Linear regression, and LASSO CV. The VADER

sentiment analysis of the tweets was positive, neutral, or negative after the thresholds were applied. And the percentage of Donald Trump and Hillary Clinton retweets were shown below.

Percentage of Trump and Clinton Retweets

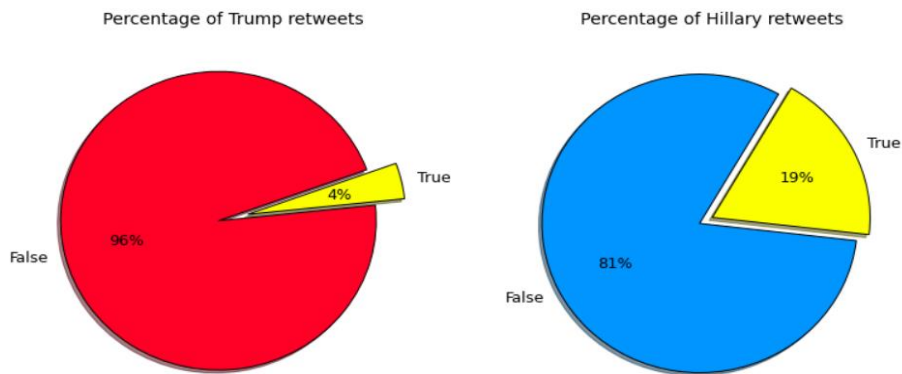


FIG 15

The machine learning algorithms like support vector regression, KNeighbor regression, Linear regression, and LASSO CV techniques were applied in this project. And the collected tweets are filtered using NLKT. Where unnecessary common words are removed. And these filtered tweets are stored in the text file for sentiment analysis to predict the opinion of people about Donald Trump and Hillary Clinton's tweets.

The suggested method is used for the filtered tweets to calculate the total number of favorable, unfavorable, and neutral tweets for Trump, and Hilary. Depending on the polarity of the specific words used in the tweet, the sentiment of the communication is categorized. Positive words (such as good, great, etc.) receive a score of +0.05, while negative words (such as terrible, worse, etc.) receive a score of -0.05, and neutral words (such as quite, average, etc.) receive a score of 0. The fraction of good, negative, and neutral tweets is then calculated using these counts. The monthly tweets of Donald Trump and Hillary Clinton counts are shown below.

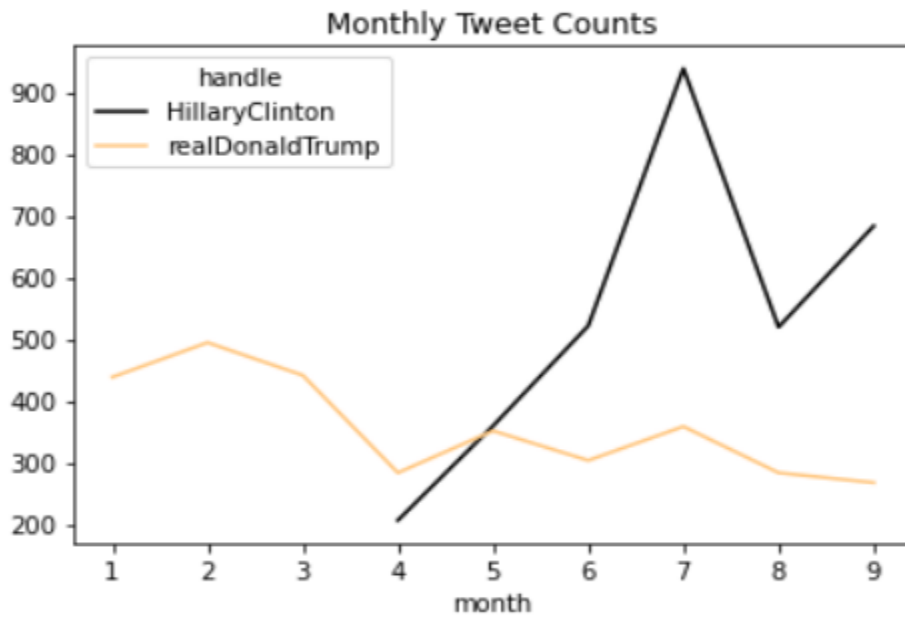


FIG 16

A word cloud Plotly provides a visual depiction of the tweets' most commonly used terms. The term that is used frequently will be prominently shown in the Word Cloud. It depicts the word cloud that was generated for Hillary by importing her edited tweets. Trump word clouds may also be found similarly.

WORD CLOUD OF DONALD TRUMP AND HILLARY CLINTON COUNTS

KNeighbor regressor

The KNN gave the best root mean squared error. A K nearest neighbor is a regressor known as a KNeighbor. It is a straightforward algorithm that predicts the numerical target. In statistics and pattern recognition, KNN is employed. Algorithm It is simple to implement KNN regression by calculating the average of the numerical objective of the K nearest neighbors. The inverse distance-weighted average of the K nearest neighbors is a different approach. In both KNN classification and KNN regression, the same distance functions are employed. The final root mean square error is 0.058065951350744924

Linear regression

One of the best machine learning algorithms is linear regression. To forecast the value of one variable based on the value of another, linear regression is utilized. The dependent variable refers to the variable we wish to forecast. The term "independent variable" refers to the variable we are utilizing to forecast the other variable. The final root mean square error is 0.036087488868269214

LASSO CV

Ridge regression is distinct from LASSO regression. It normalizes using coefficient values. The primary problem with lasso regression when we have linked variables is that it only keeps one variable and sets the other associated variables to zero. Because of the potential for information loss, our model's accuracy may suffer. A regularisation method for more precise prediction is LASSO regression. The final squared root mean squared error is 0.030372749896905225

MACHINE LEARNING ALGORITHM	ROOT MEAN SQUARE ERROR
SUPPORT VECTOR REGRESSION	0.042350035017268604.
LINEAR REGRESSION	0.036087488868269214
KNNEIGHBOR REGRESSOR	0.058065951350744924
LASSO CV	0.030372749896905225

FIG 18

The LASSOCV is the best accuracy in these machine learning algorithms which gives 0.030372749896905225. The overall performance of the machine learning algorithm is plotted in the graph chart shown below.

OVERALL PERFORMANCE OF THE MODEL

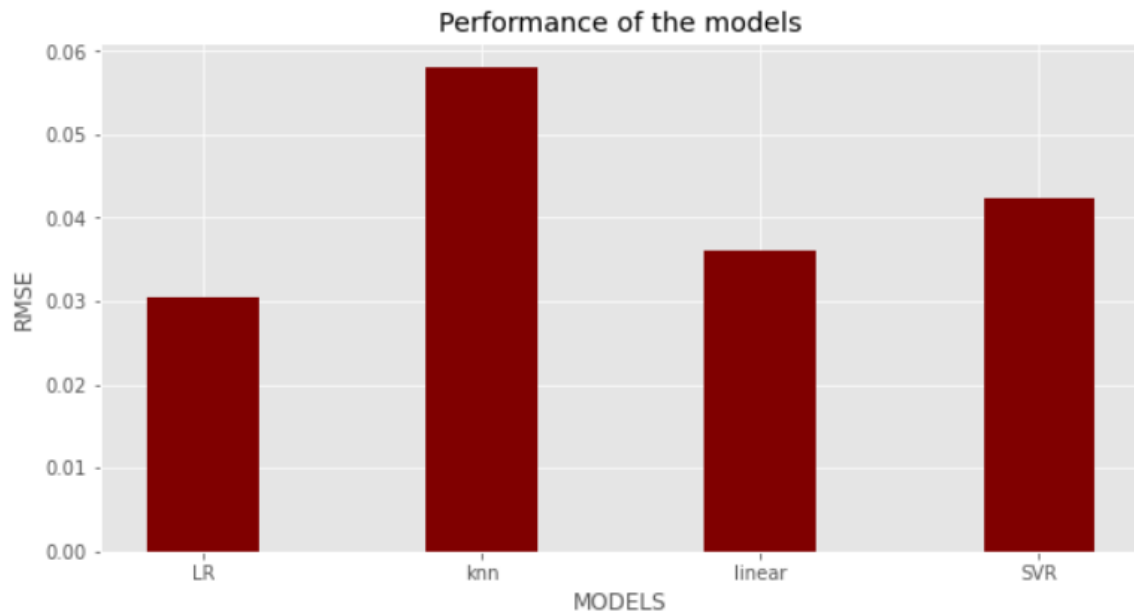


FIG 19

14. Conclusion and Future Enhancement

In this paper, we proposed VADER sentiment analysis, NLTK, and machine learning algorithms like SVR (support vector regression), Linear regression, KNeighbor regressor, and LASSO CV. In these algorithms, LASSOCV gave the best accuracy, and SVR and linear shows the good result in prediction. The tweets are considered from the US presidential election of Donald trump and Hillary Clinton. This measure signifies the positive, negative, and neutral attitudes of people toward election candidates. We can say that more cleaned data will give a good result We have to enhance the sentiment analysis method by updating the new VADER-like algorithm for the process of capturing more words in a text. And have to implement the best machine learning algorithm for the text analysis.

The future of sentiment analysis will involve digging even farther under the surface of likes, comments, and shares to properly grasp the value of social media interactions and what they reveal about the people who use the platforms. This projection also predicts that sentiment analysis will find use in larger contexts; in addition to brands, this technology will also be used by public figures, NGOs, government agencies, schools, and many other institutions. Sentiment analysis is getting better because of the increase in social media.

The majority of analytics, including social media and sentiment analysis, has long been based on algorithms. Many analytics solutions are turning to machine learning and artificial intelligence as a replacement for algorithms as these fields have made significant strides in recent years. Unfortunately, machine learning isn't yet ready to handle the complex nuances of text and how we talk, especially on social media channels that are rife with slang, sarcasm, double meanings, and misspellings. This is bad news for organizations looking to use sentiment analysis to measure audience emotions. Due to

these, it is challenging for artificial intelligence systems to correctly organize and categorize social media sentiment.

These data technologies were employed to assess people's emotions during the Brexit vote and the 2016 US presidential election to predict how these events would turn out. Due to this, businesses that are not brands are now using sentiment analysis for their purposes. Additionally, because of newly developing social media platforms and capabilities, the insights gained from these technologies are getting increasingly deeper.

APPENDIX

GITLAB CODE:

[21-22 CE901-SU / 21-22 CE901-SU_ramamoorthy_sandeep_k · GitLab \(essex.ac.uk\)](https://gitlab.essex.ac.uk/21-22_CE901-SU/21-22_CE901-SU_ramamoorthy_sandeep_k)

Reference

- [1] U. Yaqub, N. Sharma, R. Pabreja, S. A. Chun, V. Atluri, and J. Vaidya, "Location-based sentiment analyses and visualization of Twitter election data," *Digit. Gov.: Res. Pract.*, vol. 1, no. 2, pp. 1–19, 2020.
- [2] J. A. Caetano, H. S. Lima, M. F. Santos, and H. T. Marques-Neto, "Using sentiment analysis to define twitter political users' classes and their homophily during the 2016 American presidential election," *J. Internet Serv. Appl.*, vol. 9, no. 1, 2018.
- [3] F. Nausheen and S. H. Begum, "Sentiment analysis to predict election results using Python," in 2018 2nd International Conference on Inventive Systems and Control (ICISC), 2018, pp. 1259–1262.
- [4] "Social media on the campaign trail: Barack Obama and Donald Trump," content group, 11-Sep-2017. [Online]. Available: <https://contentgroup.com.au/2017/09/social-media-campaign-trail-obama-trump/>. [Accessed: 28-Aug-2022].
- [5] D. Shah, Y. Li, and A. Hadaegh, "Twitter-based sentiment analysis of each presidential candidate using long short-Term Memory," *Cscjournals.org*. [Online]. Available: <https://www.cscjournals.org/manuscript/Journals/IJCSS/Volume15/Issue3/IJCSS-1617.pdf>. [Accessed: 28-Aug-2022].
- [6] J. Song, K. T. Kim, B. Lee, S. Kim, and H. Y. Youn, "A novel classification approach based on Naïve Bayes for Twitter sentiment analysis," *KSII Trans. Internet Inf. Syst.*, vol. 11, no. 6, pp. 2996–3011, 2017.

- [7] M. Wankhade, A. C. S. Rao, and C. Kulkarni, "A survey on sentiment analysis methods, applications, and challenges," *Artif. Intell. Rev.*, 2022.
- [8] T. Pano and R. Kashef, "A complete VADER-based sentiment analysis of bitcoin (BTC) tweets during the era of COVID-19," *Big Data Cogn. Comput.*, vol. 4, no. 4, p. 33, 2020.
- [9] A. Band, "Twitter Sentiment Analysis using Vader & Tweepy," *Python in Plain English*, 06-May-2020. [Online]. Available: <https://python.plainenglish.io/twitter-sentiment-analysis-using-vader-tweepy-b2a62fba151e>. [Accessed: 28-Aug-2022].
- [10] S. E. Saad and J. Yang, "Twitter Sentiment Analysis Based on Ordinal Regression," *IEEE Access*, vol. 7, pp. 163677-163685, 2019.
- [11] A. Feizollah, S. Ainin, N. B. Anuar, N. A. B. Abdullah, and M. Hazim, "Halal Products on Twitter: Data Extraction and Sentiment Analysis Using Stack of Deep Learning Algorithms," *IEEE Access*, vol. 7, pp. 83354-83362, 2019.
- [12] Mittal, S. A. (2016). Sentiment Analysis of E-Commerce and Social Networking Sites. *IEEE*, pp. 2300-2305.
- [13] Shahare, F. F. (2017). Sentiment Analysis for the News Data Based on social media. *IEEE*, pp. 1365-1370.
- [14] ZiyuanZhao, HuiyingZhu, ZehaoXue, ZhaoLiu, JingTian, Matthew Chin HengChua, and MaofuLiu, "An image-text consistency driven multimodal sentiment analysis approach for social media", *Information Processing & Management*, vol. 56, no. 6, November 2019.
- [15] Kaur, H. J. (2015). Sentiment Analysis from social media in Crisis Situations. *IEEE*, (pp. 251-256).
- [16] SaeromPark, JaewookLee, and KyoungokKim, "Semi-supervised distributed representations of documents for sentiment analysis", *Neural Networks*, vol. 119, pp. 139-150, November 2019.
- [17] MohammadAl-Smadi, OmarQawasmeh, MahmoudAl-Ayyoub, YaserJararweh, and BrijGupta, "Deep Recurrent neural network vs. support vector machine for aspect-based sentiment July 2018.
- [18] Hyun-jungPark, MinchaeSong, and Kyung-ShikShin, "Deep learning models and datasets for aspect term sentiment classification: Implementing holistic recurrent attention on target-dependent memories", *Knowledge-Based Systems*, vol. 187, January 2020.
- [19] ParamitaRay, and AmlanChakrabarti, "A Mixed approach of Deep Learning method and Rule-Based method to improve Aspect Level Sentiment Analysis", *Applied Computing and Informatics*, Available online 4 March 2019.
- [20] NeelamMukhtar, Mohammad AbidKhan, and NadiaChiragh, "Lexicon-based approach outperforms the Supervised Machine Learning approach for Urdu Sentiment Analysis in multiple domains", *Telematics and Informatics*, vol. 35, no. 8, pp. 2173-2183, December 2018

- [21] M. Afzaal, M. Usman and A. Fong, "Tourism Mobile App With Aspect-Based Sentiment Classification Framework for Tourist Reviews," *IEEE Transactions on Consumer Electronics*, vol. 65, no. 2, pp. 233-242, May 2019.
- [22] SaeromPark, JaewookLee, and KyoungokKim, "Semi-supervised distributed representations of documents for sentiment analysis", *Neural Networks*, vol. 119, pp. 139-150, November 2019.
- [23] Shihab Elbagir and Jing Yang, "Twitter Sentiment Analysis Using Natural Language Toolkit and VADER Sentiment", *Proceedings of the International MultiConference of Engineers and Computer Scientists 2019 IMECS 2019*, March 13-15, 2019, Hong Kong
- [24] Capturing investor sentiment from Big Data: the effects of online social media on SET50 index - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Process-flow-of-the-sentiment-analysis-for-a-sentiment-index_fig1_341914247 [accessed 30 Aug 2022]
- [25] Hutto, C.J. & Gilbert, E.E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. Eighth International Conference on Weblogs and social media ICWSM-14). Ann Arbor, MI, June 2014.
- [26] Kiran Hegde. Architect, Aarush Gupta. "Sentiment Analysis in the IT Domain an Enhanced Approach to VADER Sentiment", *International Journal of computer trends and technology (1)CCT* – Volume 59 Issue 1 – May -2018.
- [27] *Datacamp.com*. [Online]. Available: <https://www.datacamp.com/tutorial/text-analytics-beginners-nltk>. [Accessed: 30-Aug-2022].
- [28] "Tokenization in NLP," *Kaggle.com*, 06-Dec-2020. [Online]. Available: <https://www.kaggle.com/code/satishgunjal/tokenization-in-nlp>. [Accessed: 30-Aug-2022].
- [29] P. Pdamkar, "Support Vector Regression," *EDUCBA*, 06-Feb-2020. [Online]. Available: <https://www.educba.com/support-vector-regression/>. [Accessed: 30-Aug-2022].
- [30] "ML," *GeeksforGeeks*, 13-Sep-2018. [Online]. Available: <https://www.geeksforgeeks.org/ml-linear-regression/>. [Accessed: 30-Aug-2022].
- [31] I. Muhajir, "K-neighbors regression analysis in python," *Analytics Vidhya*, 20-Apr-2019. [Online]. Available: <https://medium.com/analytics-vidhya/k-neighbors-regression-analysis-in-python-61532d56d8e4>. [Accessed: 30-Aug-2022].
- [32] "A Complete understanding of LASSO Regression," *GreatLearning Blog: Free Resources what Matters to shape your Career!*, 26-Dec-2021. [Online]. Available: <https://www.mygreatlearning.com/blog/understanding-of-lasso-regression/>. [Accessed: 30-Aug-2022].